

BUCKLING OF COMPOSITE CURVED SHELLS

Part: **3.4 – STRENGTH OF COMPOSITE PARTS**

Chapter: **3455**

Title: **BUCKLING OF COMPOSITE CURVED SHELLS**

Revision: **0.1**

Date: **01/10/2021**

1 INTRODUCTION

1.1 OBJECT

This chapter describes the methodology proposed for calculating the buckling onset of **rectangular shells (of single curvature)** made of composite laminates. Two sets of methods are presented:

✓ Pre-sizing methods

To be used in initial phases of the design process. They are based in exact solutions or approximations found in literature, engineering handbooks, or in internal Airbus DS development in some cases.

✓ Methods for final strength check

To be used in final sizing or in certification strength justification. They are based in numerical solutions, have a higher computational cost and shall be utilized only using validated software implementations.

1.2 SCOPE AND LIMITATIONS

Buckling of composite **shells**. Limitation of pre-sizing methods:

- ✓ Rectangular shape
- ✓ **Single curvature shell**, simply supported on all its perimeter
- ✓ Other limitations are listed in the dedicated sub-chapter

In any case, the post-buckling behavior of the **shell** shall not be discussed here.

BUCKLING OF COMPOSITE CURVED SHELLS

2 METHODS FOR PRE-SIZING

The pre-sizing methods consist of analytical solutions, handbook methods and available COTS software (such as ESDU programs) with specific simplifications and resulting advantages and disadvantages.

2.1 CYLINDRICAL NARROW SHELLS

Shells under consideration within this section have rectangular shape, single curvature, are simply supported in its perimeter, its axial length greater than its width (the arc length in the circumferential direction), and that arc length is small compared with the curvature radius of the shell. This shell model can be representative of skin panels limited by stringers and frames in conventional fuselages.

Compression buckling of composite shells

From ESDU94007 the solution of the buckling equation for a cylindrical shell, made of an orthotropic composite laminate ($B_{ij} = D_{16} = D_{26} = 0$)(*), under bi-axial loads is:

$$N_{xb} = - [(H_{12}^2 / H_{11}) + H_{22} + N_{yb} \cdot \eta^2] / \lambda^2$$

$$H_{11} = \lambda^4 \cdot S_{22} + 2 \cdot [S_{12} + (S_{33}/2)] \cdot \lambda^2 \cdot \eta^2 + S_{11} \cdot \eta^4$$

$$H_{12} = \lambda^2 / R$$

$$H_{22} = \lambda^4 \cdot D_{11} + 2 \cdot (D_{12} + 2 \cdot D_{66}) \cdot \lambda^2 \cdot \eta^2 + D_{22} \cdot \eta^4$$

$$[S] = [A]^{-1}$$

$$\lambda = m \cdot \pi / a$$

$$\eta = n \cdot \pi / b$$

... where:

- a , b are respectively the length (X-direction) and the width (Y/arc direction) of the shell
- R is the curvature radius of the shell (around an axis parallel to X direction)
- m , n are the number of buckling semi-waves in longitudinal and transversal directions
- N_{xb} , N_{yb} are the loads at the buckling onset (negative values if compression)
- D_{11} , D_{12} , D_{22} , D_{66} are the non-zero terms of the flexural stiffness matrix of the laminate

(*) See discussion about specially orthotropic plates and the generalized bending stiffness matrix of laminates in section 2.1 of chapter 3450 "BUCKLING OF COMPOSITE PLATES"

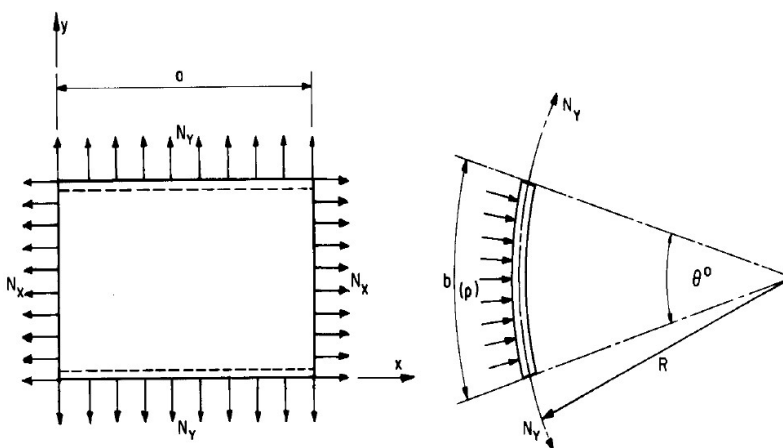


Figure 1. Shell geometry and loads in ESDU94007

BUCKLING OF COMPOSITE CURVED SHELLS

If buckling load is expressed in terms of times the applied loads, i.e. the Buckling Level (equivalent to Reserve Factor if buckling were considered failure)...

$$N_{xb} = BL \cdot N_x, \quad N_{yb} = BL \cdot N_y$$

... the solution becomes:

$$BL = - [(H_{12}^2 / H_{11} + H_{22}) / \lambda^2] / [N_x + (N_y \cdot \eta^2) / \lambda^2]$$

The critical buckling load solution shall be found by iterating over the possible number of semi-waves (m, n from 1 to ...) to find the minimum value.

Shear buckling of composite shells

No analytical solution or approximation has been found in literature for the buckling shear load of a composite shell. The proposed method shall start by calculating the shear buckling load of the rectangular plane plate with the same length, width and laminate of the original curved shell. Later on, the curvature correction shall be made using one of the following methods...

- The same method included in EADS Defence & Security multidisciplinary optimization tool LAGRANGE, proposed by Wiedemann:

$$N_{xyb} = N_{xyb\text{-flat}} \cdot (1 + \Omega / 700)^{1/4}$$

$$\dots \text{ where } \quad \Omega = \text{Min} \{ 40^2, 12 \cdot (1 - \nu_{xy} \cdot \nu_{yx}) \cdot [b^2 / (R \cdot t)]^2 \}$$

- A similar approach to account the curvature in shear buckling can be done approximating a representative population of design configurations, calculated by means of FEM, to a simple function of the curvature parameter (typical expressions for this parameter "Ω" in previous page, or "Z" in Niu's book, §11.4, or just $b^2 / (R \cdot t)$ in this case). Proposed design model is:

$$N_{xyb} = N_{xyb\text{-flat}} \cdot [1 + k \cdot b^2 / (R \cdot t)]$$

$$\dots \text{ where } \quad k = 0.024 + 0.01 \cdot (a/b - 1)^{0.5}$$

...if $b > a$ use $a/b = 1$ and replace b by a in previous formulas

The representative population considered consisted of combinations of 4 slenderness ratios (a/b), 4 curvature radius (R equal to 500, 1000, 2000 and 3000mm), 3 stringer pitches (b equal to 150, 200 and 300mm) and 10 laminates with different stacking styles and thicknesses from 1.5 up to 7mm. Almost 500 runs of MSC-Nastran buckling solution 105 were used.

BUCKLING OF COMPOSITE CURVED SHELLS

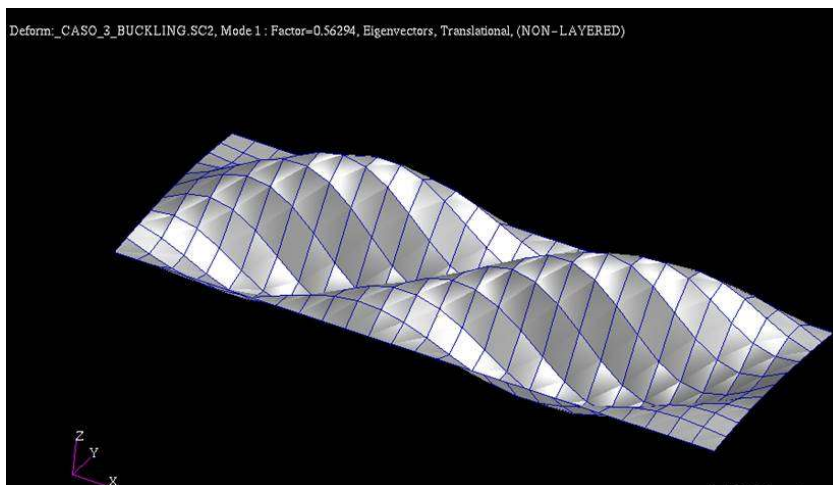


Figure 2. Example of FEM mesh used to build the approximation with Nastran SOL 105

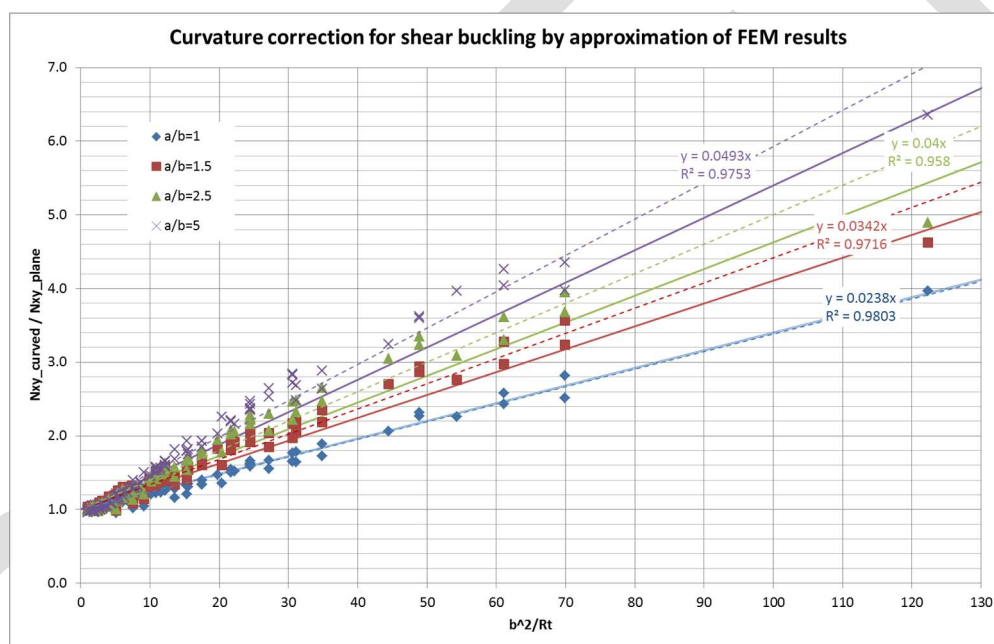


Figure 3. Shear buckling FEM results, trend lines (dashed lines) and design model (cont. lines)

2.2 CYLINDRICAL BARRELS (OR WIDE SHELLS)

Under this definition, shells will have a width significantly greater than the axial length, and typically with an arc length of the order of the curvature radius or greater. The analysis methods proposed for this type of shell can be considered representative of typical fuselages of combat aircrafts (with few or no longitudinal stringers and frames at shorter pitch).

Buckling of composite cylinders in axial compression

K.F. Wilson Wong and Paul M. Weaver studied the buckling of cylinders made of laminated composites under compression axial load (Ref. 19), arriving to a closed form solution, valid even for non-orthotropic laminates with bending-extension and bending-torsion couplings.

$$P = \frac{1}{\lambda_m^2} \left[d_{11}^* \lambda_m^4 - 4d_{16}^* \lambda_m^3 \lambda_n + 2(d_{12}^* + 2d_{66}^*) \lambda_m^2 \lambda_n^2 - 4d_{26}^* \lambda_m \lambda_n^3 + d_{22}^* \lambda_n^4 \right] + \frac{\left[b_{21} \lambda_m^4 + (b_{61} - 2b_{26}) \lambda_m^3 \lambda_n + (b_{11} + b_{22} - 2b_{66}) \lambda_m^2 \lambda_n^2 + (b_{62} - 2b_{16}) \lambda_m \lambda_n^3 + b_{12} \lambda_n^4 + \lambda_m^2 / r \right]^2}{\lambda_m^2 \left[a_{22} \lambda_m^4 + 2a_{26} \lambda_m^3 \lambda_n + (2a_{12} + a_{66}) \lambda_m^2 \lambda_n^2 + 2a_{16} \lambda_m \lambda_n^3 + a_{11} \lambda_n^4 \right]}$$

Figure 4. Buckling load of a laminated circular cylindrical shell under compression, per Wilson and Weaver (eq. [12], Ref. 19)

Where:

- a_{ij} terms of the membrane compliance matrix $[A]^{-1}$ of the laminate
- b_{ij} terms of the eccentricity matrix $[A]^{-1} [B]$
- d_{ij}^* modified/generalized bending stiffness matrix
- r radius of the cylinder
- λ_n circumferential wavelength, equal to n/r (n positive integer number)
- λ_m axial wavelength, equal to $m' \cdot n/r$ (m' slope of spiral, any real number)

Finding the solution P above (force per unit length, equivalent to the symbols $N_{x,cr}$ or N_{xb} in other sections of this document) requires a search of the minimum along the possible values of variables λ_m and λ_n . Simplifications done for obtaining that formula require compliance with the following conditions:

$$\text{Batdorf parameter} \quad l^2/(r \cdot t) > 100 \qquad \text{Length parameter} \quad l^2 \cdot t/r^3 < 3$$

The authors found good correlation between the predictions resulting from this formula and the correspondent solutions obtained by means of FEM analysis. This method shall be the recommended one for wide cylindrical composite shells subjected to axial compression load distributions, either uniform across the width, or with a moderate variation (eg. that due to gross bending of the cylinder). In the later case, the reference load to compare with the buckling allowable derived by the method, when obtaining static margins, shall be the maximum axial compression load found in the shell.

BUCKLING OF COMPOSITE CURVED SHELLS

Buckling of composite cylinders (NASA SP-8007)

The NASA proposed a closed form methodology for calculating the buckling onset of thin-walled cylinders made of isotropic or composite materials, with might include also stiffeners (Ref. 16, discussed with a more familiar notation nowadays in Ref. 17 or Ref. 18). This methodology would be applicable also to wide shells, i.e. those with the circumferential dimension much greater than the axial direction (provided no other intrinsic limitations of the method is violated).

- ✓ Axial compression buckling

$$N_{x,cr} = \frac{1}{\alpha^2} \frac{\begin{vmatrix} C_{11} & C_{12} & C_{13} \\ C_{12} & C_{22} & C_{23} \\ C_{13} & C_{23} & C_{33} \end{vmatrix}}{\begin{vmatrix} C_{11} & C_{12} \\ C_{12} & C_{22} \end{vmatrix}}$$

Where:

$$\alpha = m\pi/L$$

$$\beta = m\pi/R$$

$$C_{11} = A_{11} \alpha^2 + A_{66} \beta^2$$

$$C_{12} = (A_{12} + A_{66}) \alpha \beta$$

$$C_{13} = A_{12} \alpha / R + B_{11} \alpha^3 + B_{12} \alpha \beta^2 + 2 B_{66} \alpha \beta^2$$

$$C_{22} = A_{22} \beta^2 + A_{66} \alpha^2$$

$$C_{23} = B_{22} \beta^3 + B_{12} \alpha^2 \beta + 2 B_{66} \alpha^2 \beta + A_{22} \beta / R$$

$$C_{33} = D_{11} \alpha^4 + 2 (D_{12} + 2 D_{66}) \alpha^2 \beta^2 + D_{22} \beta^4 + (A_{22} / R + 2 B_{22} \beta^2 + 2 B_{12} \alpha^2) / R$$

The terms of the $[A]$, $[B]$ and $[D]$ stiffness matrices of the laminate appear as factors in equations above. Iterating along m and n (number of buckling waves in axial and circumferential directions), the minimum value for $N_{x,cr}$ shall be the buckling onset.

- ✓ Hoop compression buckling (external pressure)

$$p_{cr} = \frac{R}{n^2} \frac{\begin{vmatrix} C_{11} & C_{12} & C_{13} \\ C_{12} & C_{22} & C_{23} \\ C_{13} & C_{23} & C_{33} \end{vmatrix}}{\begin{vmatrix} C_{11} & C_{12} \\ C_{12} & C_{22} \end{vmatrix}}$$

The terms of the $[C]$ matrix are the same as in previous case, the iterative process is again required, and the relation between the critical external pressure and the buckling onset in hoop compression is simply:

$$N_{y,cr} = p_{cr} R$$

- ✓ Torsion

$$T_{cr} = 21.75 (D_{22})^{5/8} \left(\frac{A_{11}A_{22} - A_{12}^2}{A_{22}} \right)^{3/8} \frac{R^{5/4}}{L^{1/2}}$$

BUCKLING OF COMPOSITE CURVED SHELLS

In principle valid only for moderately long and long cylinders. Note that the buckling shear load in the shell can be obtained from that critical torsion moment in the cylinder as:

$$N_{xy,cr} = T_{cr} / 2\pi R^2$$

CORRELATION FACTOR AND SIMPLIFIED AXIAL BUCKLING FOR UNSTIFFENED CIL.?

KDF = 0.75 if stiffeners?

DRAFT

BUCKLING OF COMPOSITE CURVED SHELLS

2.3 BUCKLING UNDER COMBINED LOADS

Buckling under combined loads in narrow shells

The same approach proposed for flat rectangular plates in section 2.5 of chapter 3450 "BUCKLING OF COMPOSITE PLATES" shall be followed.

Buckling under combined loads in barrels and wide shells

References consulted propose to use the (most conservative) linear interaction law to find the buckling level in combined loads from the buckling ratios in uni-axial loads (see Ref. 16, Ref. 17 or Ref. 18, axial compression and lateral pressure, or axial compression and torsion). Thus, buckling will occur if...

$$R_c + R_p + R_s = 1$$

Where:

- R_c is the buckling ratio in axial loads, equal to $-N_x/N_{xb}$ if $N_x < 0$ (compression), or 0 otherwise (tension)
- R_p is the buckling ratio in circumferential loads, equal to $-N_y/N_{yb}$ if $N_y < 0$ (compression, external pressure), or 0 otherwise (tension, internal pressure)
- R_s is the buckling ratio in torsion/shear load, equal to $|N_{xy}/N_{xyb}|$

For finding the static margin associated to a particular load set, the expression for the reserve factor would be:

$$RF = 1 / (R_c + R_p + R_s)$$

As discussed in the introductory section, the post-buckling behavior of the shell is not discussed in this chapter. Nevertheless, it is worth to remark that, in the case of wide (or fully cylindrical) shells, there will be not longitudinal stringers capable to take the overloads sent to its frame by a post-buckled skin (see discussions about the effective skin approach or the diagonal tension theory in other chapters), and then, it is not reasonable to expect a significant post-buckling capability.

3 METHODS FOR FINAL STRENGTH CHECK

The methods presented in previous section are generally, conservative, and indeed may still be used in final sizing steps, or in certification justification, provided their limitations have not been exceeded. In addition to those inherent limitations of the analytical methods, under some circumstances, more refined methodologies shall be needed to evaluate the accurately the buckling levels of the part. These circumstances may include short static margins, weight saving campaigns, loads exceedances in final load loops, ..., among others.

BUCKLING OF COMPOSITE CURVED SHELLS**Detailed Finite Elements Models**

For specially complicated parts the use of FEM analysis is the last and more sophisticated tool to solve the buckling analysis problem.

Rules and recommendations for adequate preparation and processing FEM analyses are out of the scope of this chapter. See Volume 2 of the Stress Methods Manual for further information.

DRAFT

BUCKLING OF COMPOSITE CURVED SHELLS

4 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
a	mm	Length of the plate/shell (dimension in X-longitudinal direction)
$[A]$	N/mm	In-plane stiffness matrix of a composite laminate
b	mm / -	Width of the plate/shell (dimension in Y-transversal direction) or, as subscript, relative to buckling
c	-	As subscript, relative to compression
CRFP	-	Carbon Fiber Reinforced Polymer
COTS	-	Commercial off-the-shelf
$[B]$	N	Matrix defining the laminate shear-bending coupling
BL	-	Buckling level, ratio between buckling load and applied load
$[D]$	Nmm	Flexural stiffness matrix of a composite laminate
f	-	As subscript, relative to in-plane bending
I	mm ⁴	Bending moment of inertia
M_b	Nmm	In-plane bending moment of the plate at buckling onset
N_{xb}	N/mm	Compression buckling load-flow (force per unit width) in X direction
N_{xyb}	N/mm	Shear buckling load-flow (force per unit width)
N_{yb}	N/mm	Compression buckling load-flow (force per unit width) in Y direction
q	N/mm	In plane shear force per unit width
r, R	mm / -	Curvature radius of the shell (around an axis parallel to X direction) or, from context, if written in upper-case, failure index, i.e. ratio between uni-axial load and allowable load
RF	-	Reserve Factor, ratio between allowable load and applied load
s / xy	-	As subscripts, relative to in-plane shear
t	mm	Thickness of the plate
τ	N/mm ²	In-plane mean shear stress
UD	-	Unidirectional

BUCKLING OF COMPOSITE CURVED SHELLS

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[S. G. Lekhnitskii]	1984	Anisotropic Plates
Ref. 2	[SAE] CMH-17	2012	Composite Materials Handbook (Volume 3)
Ref. 4	ESDU 94007	-	ELASTIC BUCKLING OF CYLINDRICALLY CURVED LAMINATED FIBRE REINFORCED COMPOSITE PANELS WITH ALL EDGES SIMPLY-SUPPORTED UNDER BIAXIAL LOADING
Ref. 6	[EADS Defence & Security]	<i>DRAFT</i>	Stress criteria for global sizing in MAS LAGRANGE
Ref. 15	NT-T-ADS-97013	Is. A	Manual Teórico del Programa ARPA, versión 2.0
Ref. 16	[NASA] SP-8007	1968	BUCKLING OF THIN-WALLED CIRCULAR CYLINDERS
Ref. 17	[Vinson]	1933	Buckling of Composite Material Shells. Solid Mechanics and Its Applications (Chapter 20, Buckling of Composite Material Shells)
Ref. 18	[Mahdy et al]	2020	Buckling and stress-competitive failure analyses of composite laminated cylindrical shell under axial compression and torsional loads
Ref. 19	[K.F.Wilson, P.M. Weaver]	2005	Approximate Solution for the Compression Buckling of Fully Anisotropic Cylindrical Shells

BUCKLING OF COMPOSITE CURVED SHELLS

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>
NASTRAN (MSC-Software)	>2005	Multidisciplinary structural analysis software based in the Finite Elements method

Table 1. Software programs referenced in this chapter

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
0.1 30/09/2021	G. Baños F. Glock	First issue	All pages

Table 2. Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
Stress Methods (TEPAP-TL4)	G. Baños 30/09/2021	Florian Glock XX/09/2021

Table 3. Approval signatures table for last revision of this chapter